

Element A: Problem Statement

Our team noticed that the majority of the Roosevelt population complained about the lack of windows in our building. Therefore, our team decided to go to one of the teachers suffering from this lack of windows (Mr. Patrick Gleason) and see what he and his students thought about the issue. According to Mr. Gleason, the lack of windows in the classroom leads to a dark and dreary place of learning. In addition, further research (from the [American Optometric Association](#)) showed that a lack of windows can lead to the following: eyestrain, headaches, blurred vision, an increase in SAD (Seasonal Affective Disorder), and a lack of creativity in students. All of this would disturb the learning environment for any class but for an English class like Mr. Gleason's, being able to feel creative and work to your best ability is very important for a student's success.

After doing further research, the lack of windows isn't just an ERHS problem. It's a nationwide problem affecting students in 32% of schools based on a report in "[Condition of America's Public School Facilities](#)".

Schools recall feeling "[a little bit like being in a box.](#)" In ERHS, this effect is observed in several classrooms, such as Dr. Yarbough's room (006) and Mrs. Dortch's room (207) both lack windows and are mentally unstimulating to be in—a stark difference to classrooms like Mr. Paz's (008) and Mr. Hawk's (015) which are abundant in windows and are nice to be in. Not only does this problem affect student health, but also academic standing. Previous studies by the [National Bureau of Economic Research](#) indicate that high school students usually score lower on tests when faced with suffocating temperatures and air quality in their classrooms. According to [CNN](#), many areas in the Maryland area are over 41 years old, and more than a third are over 51 years old. This can affect the functionality of these buildings, including the overall structure and installation of windows.

The stakeholders include (1) Mr. Gleason, (2) Mr. Gleason's students who suffer from the lack of sunlight, (3) Mr. Paz, the engineering teacher as he is the supervisor of the project, and (5) the mentor

who will assist in building the project, (6) and other students and staff in the building who are facing the same problem. By interviewing Mr. Gleason and his students, we learned that in order to sufficiently solve the problem, the considerations include the quality, stability, and outcome of the product within a respected time frame. Furthermore, by talking to Mr. Paz and the group members, we learned that the main constraints would be time, building materials, and material cost. Considering all of these factors our solution became obvious- to effectively solve the problem, we would need to build a fake window for Mr. Gleason's classroom. This window would light up, have a nature-themed background, be 3D, and solve some of the problems that a lack of actual windows caused.

Our team believes that the lack of windows is an important issue to solve because of the sheer magnitude of the issue. Not only does it affect 95% of Roosevelt classrooms, meaning 100% of Roosevelt students, but it also affects students nationwide. The effects on students' health are far-reaching, impacting their mental, social, physical, and academic health. With an artificial window, it can reap the benefits that come with having a real one without negatively impacting the school building. This can soon help all schools across the country, making sure that all students are mentally prepared to work indoors for long hours without getting exhausted.

Sources:

- <https://www.aoa.org/healthy-eyes/eye-and-vision-conditions/computer-vision-syndrome?sso=y#:~:text=The%20most%20common%20symptoms%20associated,Blurred%20vision.>
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